



AI – DRIVEN SMART SURVEILLANCE FOR WEAPON DETECTION USING YOLOv8

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ABSTRACT

Security risks caused by firearms carried either openly or concealed demand more advanced intelligent surveillance systems. The research presents a weapon identification system operating in real time and the Django-based server handles storage, together with OpenCV image processing and YOLOv8 object detection capabilities. The system collects video frames as its input and performs deep learning analyses to detect weapons in the frames as they are received in real time. The suggested method delivers fast weapon identification alerts to enhance security surveillance capabilities alongside faster response capabilities for threats that help protect public safety and law enforcement.



INTRODUCTION

Technologies Used:

Category	Technology	How it Helps
Models	YOLOv8	Detects weapons in real-time for security.
Libraries	OpenCV	Enhances video frames for better detection accuracy.
Frameworks	Django	Manages real-time monitoring and weapon detection requests.
Programming Languages	Python	Development, machine learning, and image processing.
Datasets	Custom Weapon Detection Datasets annotated on Roboflow.	Labeled datasets for training and testing weapon detection.



OBJECTIVE AND SCOPE OF THE PROJECT

Objectives:

- Develop a real-time weapon detection system using YOLOv8 and OpenCV.
- Provide **instant alerts** to security personnel upon detecting weapons.
- Enhance public safety and reduce response time to potential threats.

Scope:

- Focus on detecting weapons in video streams from surveillance cameras.
 - The system is limited to identifying weapons and not other security threats such as explosives, mines etc.
 - Deployed in public spaces like airports, malls, and stadiums.
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OBJECTIVE AND SCOPE OF THE PROJECT

Potential Applications:

- Public Security: Airports, shopping malls, public events.
- Law Enforcement: Assists police in monitoring high-risk areas.
- Smart Cities: Integration into smart city infrastructure for enhanced safety.



LITERATURE REVIEW

Summary of Existing Research/Studies:

Research Paper	Authors	Model	Findings	Limitations
Application of Deep Learning for Weapons Detection in Surveillance Videos	Tufail Sajjad Shah Hashmi et al.	YOLOv3 & YOLOv4	YOLOv4 outperforms YOLOv3 in sensitivity and processing time.	No real-time alert system, no storage.
Real-time Weapon Detection using Drone	Deepak R. Hawale, Pavin S. Game.	YOLOv3 with OpenCV	Real-time detection with drone-mounted cameras.	Relies on internet connectivity, no storage.
Weapon Detection from Images using YOLO and OpenCV	Divyanshi Chitravanshi et al.	YOLOv5 with Transfer Learning	Improved detection accuracy with YOLOv5.	Lacks storage and alert system.

RESEARCH GAP

1. High Computational Demands

Gap:

- Systems need powerful hardware, limiting deployment in resource-constrained environments.

Solution:

- Efficient YOLOv8 Model that **works on less powerful devices**.

2. Limited Scalability:

Gap:

- Systems struggle to scale for larger environments like airports or stadiums.

Solution:

- Django enables for easy scaling across multiple locations.

RESEARCH GAP

3. Complex User Interfaces

Gap:

- Systems have complicated dashboards, making quick decision-making difficult.

Solution:

- Simplified Web Interface for easy access to live feeds, alerts, and logs.



IMPORTANCE OF THE PROJECT

Need: Rising firearm-related threats require efficient weapon detection systems for **faster response** and improved safety.

Real-world Applications:

- Enhances **public safety** in places like airports, malls, and schools.
- Supports **law enforcement** in quick identification of potential threats.
- Improves **private security** response times.

Industry/Society Potential:

- Transforms the **security industry** with AI-driven real-time detection.
- Contributes to **smart cities** with automated surveillance.
- Offers global impact by reducing weapon-related violence and enhancing public security.

PROPOSED SYSTEM

Proposed Solution:

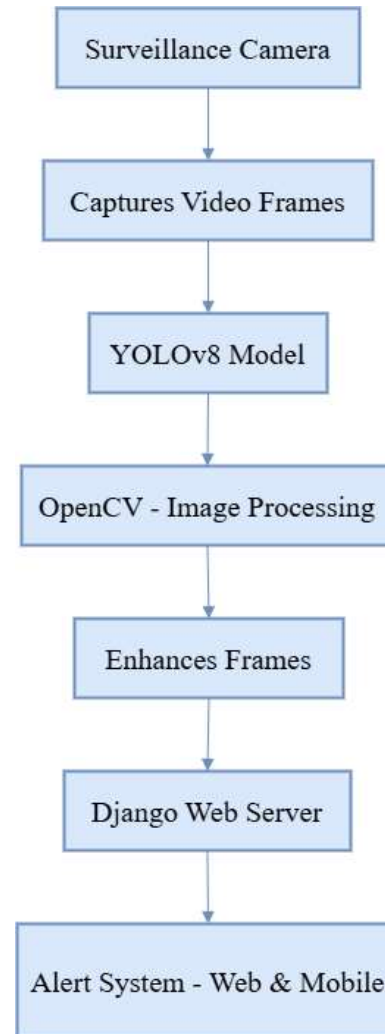
- A Real-Time Weapon Detection System using YOLOv8 for object detection, OpenCV for image processing, and a Django server for storage.
- Detected frames for efficient data management and triggers instant alerts for rapid security response.

Features & Functionalities:

- Real-time detection of **handguns**.
- Deep learning with YOLOv8 and OpenCV for image processing.
- **Instant alerts** to security personnel upon weapon detection.

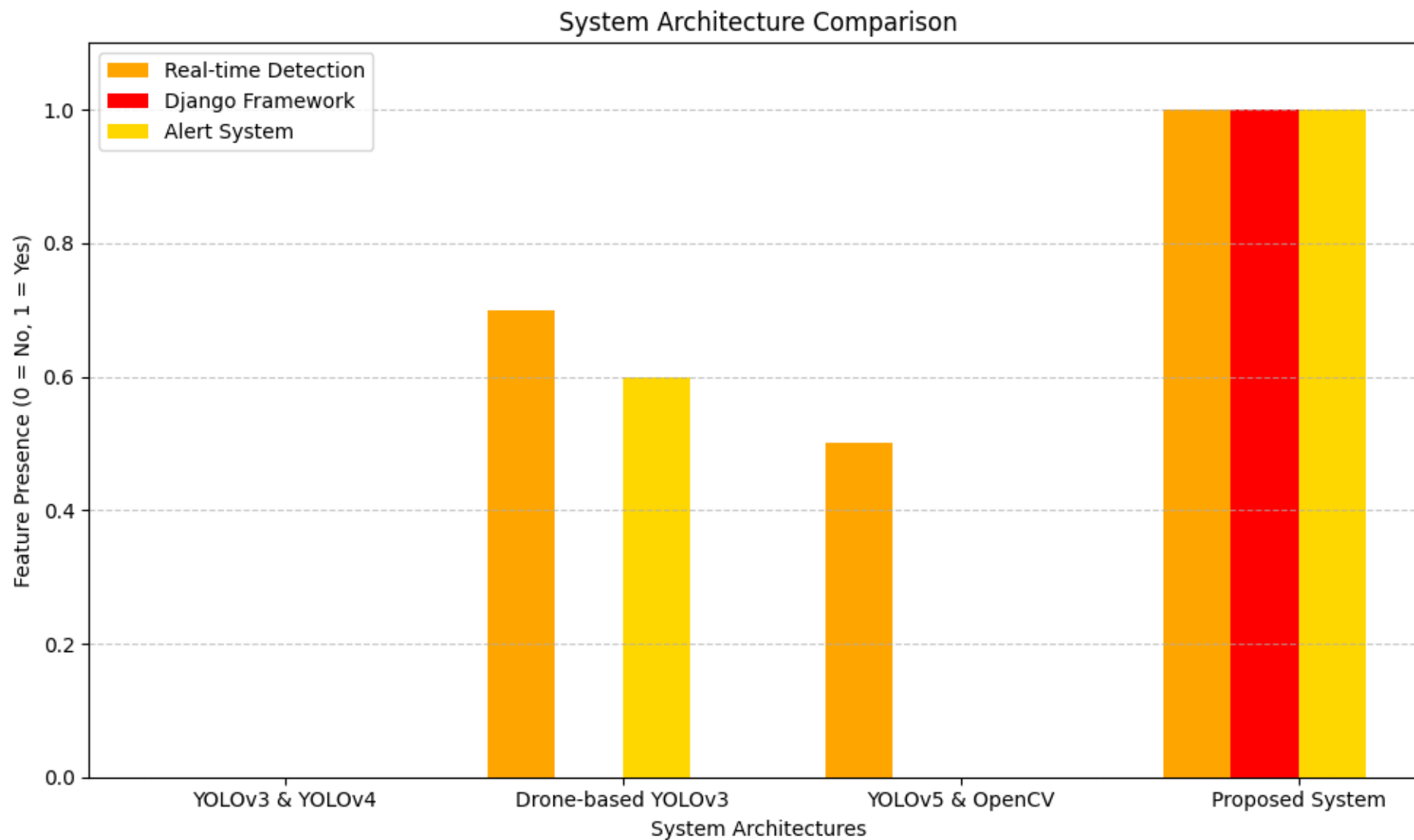
PROPOSED SYSTEM

Proposed System Architecture:



PROPOSED SYSTEM

System Architecture Comparison with previous versions:



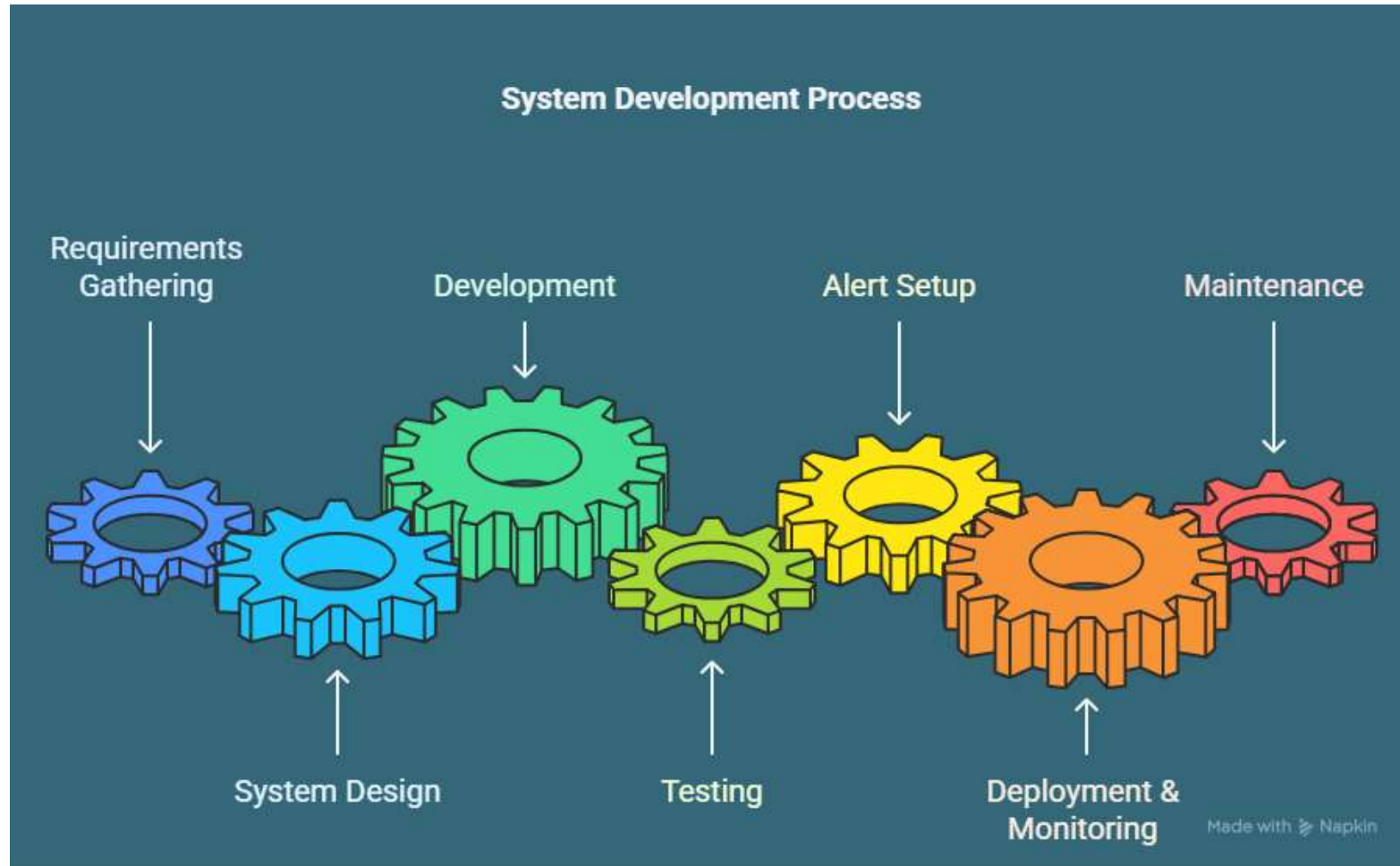
IMPLEMENTATION METHODOLOGY

Step-by-Step Execution:

1. **Requirements Gathering:** Define system needs (camera, hardware, software) and research technologies (**YOLOv8, OpenCV, Django**).
2. **System Design:** Design architecture, integrate YOLOv8, image processing.
3. **Development:**
 - Integrate YOLOv8 for detection.
 - Implement OpenCV for image enhancement.
4. **Testing:** Integrate and test system components, validate detection accuracy, and test scalability.
5. **Alert Setup:** Develop instant alert system for security personnel (**SMS or E-mail**).

6. Deployment & Monitoring: Deploy in target environments and monitor performance.

7. Maintenance: Continuously improve system accuracy and performance.



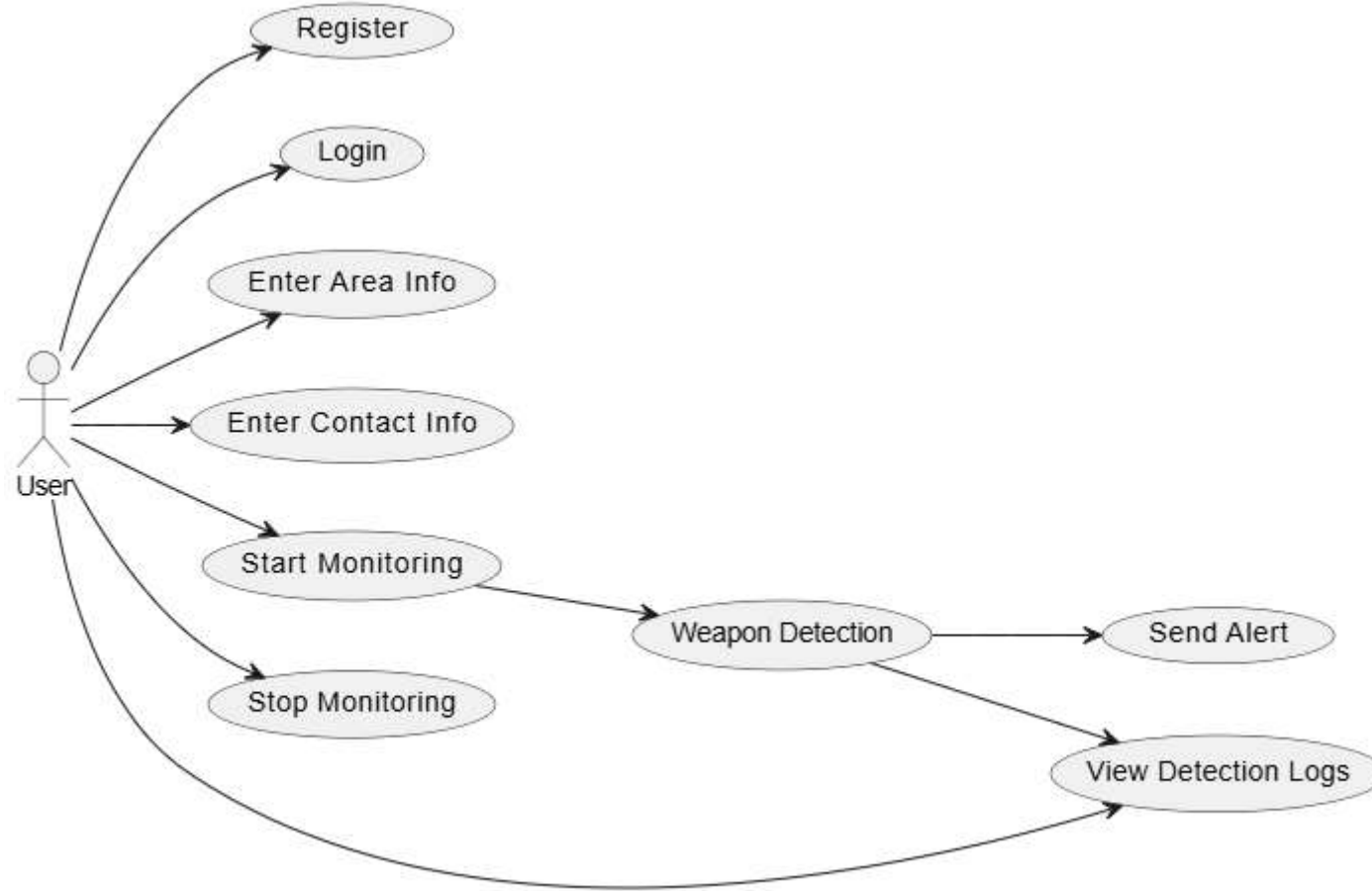
TECHNOLOGY SUPPORT FOR EXECUTION

Category	Technology	Reason for choice
Model	YOLOV8	Real-time weapon detection
Libraries	OpenCV	Image processing and enhancement
Frameworks	Django	Backend framework for deployment and alert management
Programming Languages	Python	Ideal for AI and image processing with robust libraries
Datasets	Weapon Detection Datasets	Labeled datasets for training and testing



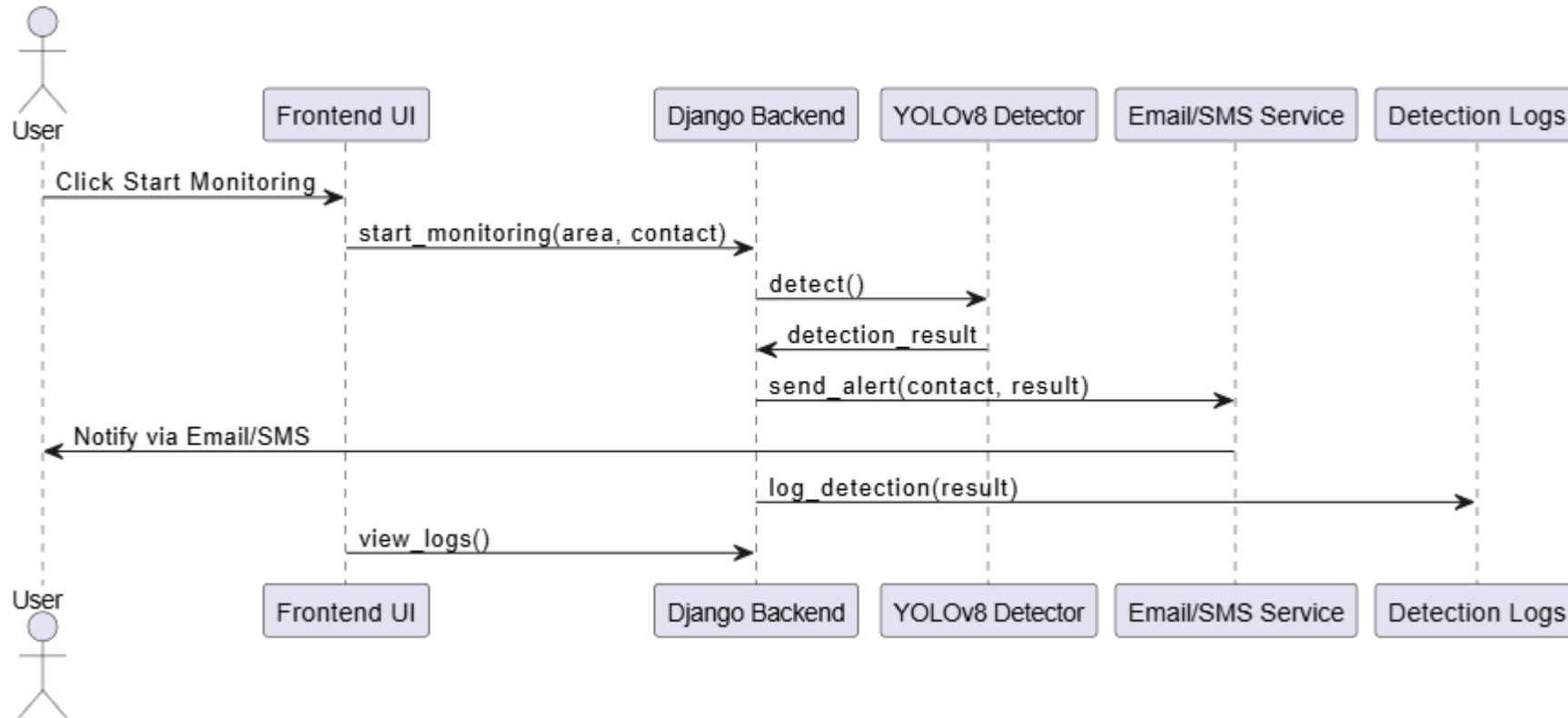
UML Diagrams:

Use Case Diagram:



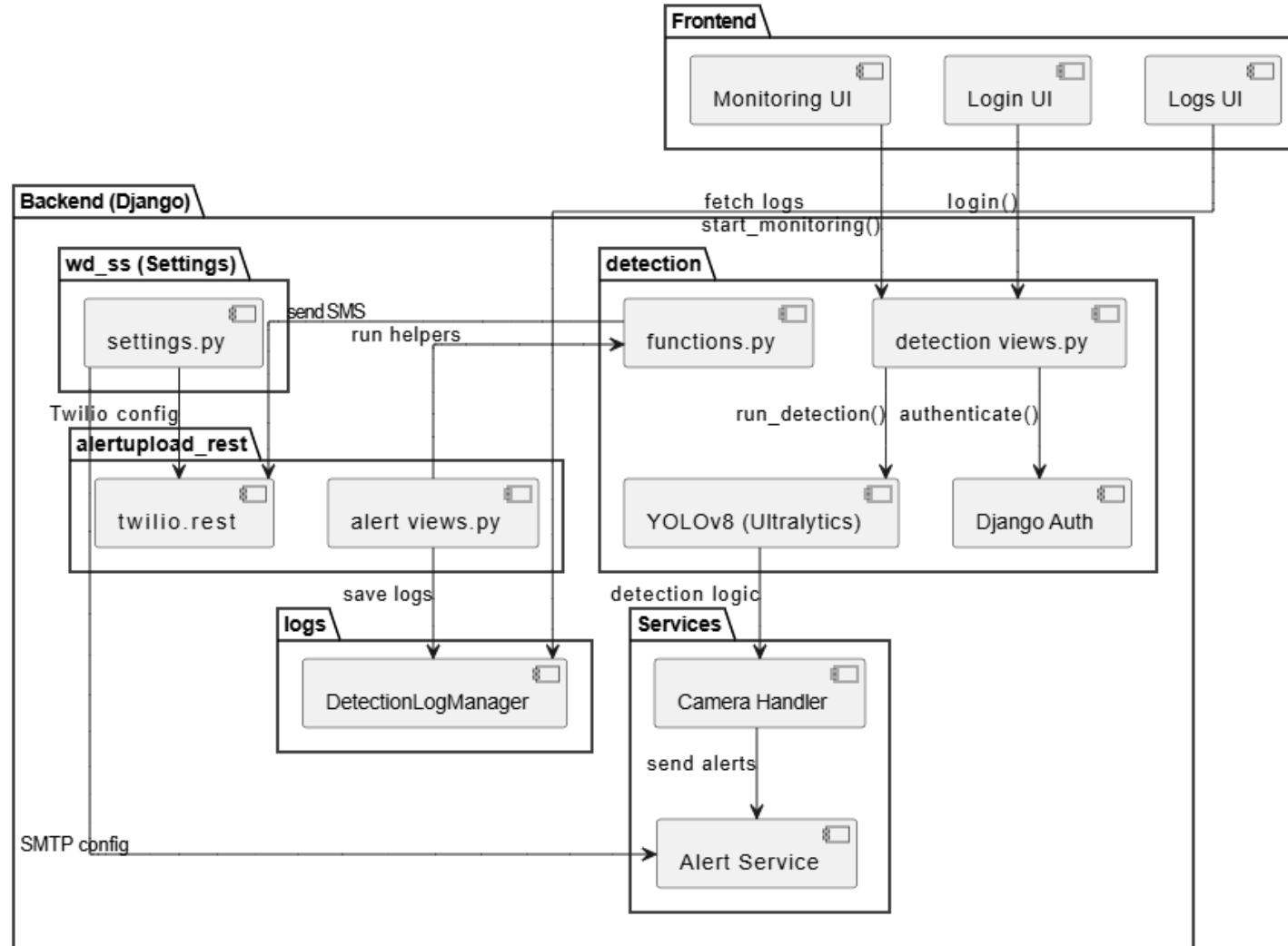
UML Diagrams:

Sequence Diagram:



UML Diagrams:

Package Diagram:



UTP & ITP

UTP – Unit Test Plan

Test Case ID	Module	Test Description	Input	Expected Output	Result
UTP-01	YOLOv8 Detection	Detect weapon in a given image	Image with gun	Weapon Detected	Pass
UTP-02	Email Alert Module	Verify email is triggered	Detection event	Email sent to recipient	Pass
UTP-03	Camera Feed	Verify webcam captures frames correctly	Webcam access	Frame stream generated	Pass

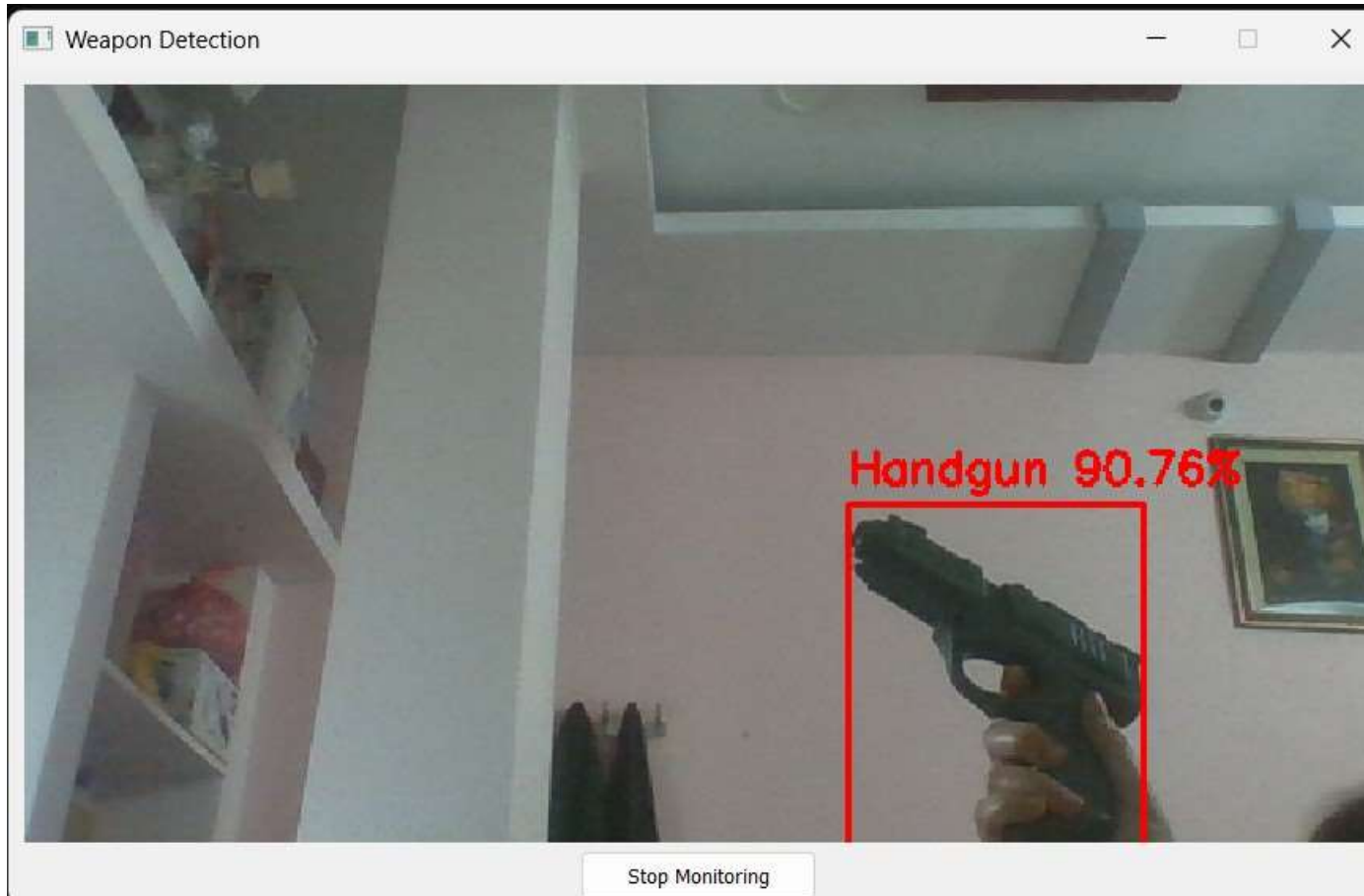


UTP & ITP

ITP – Integration Test Plan

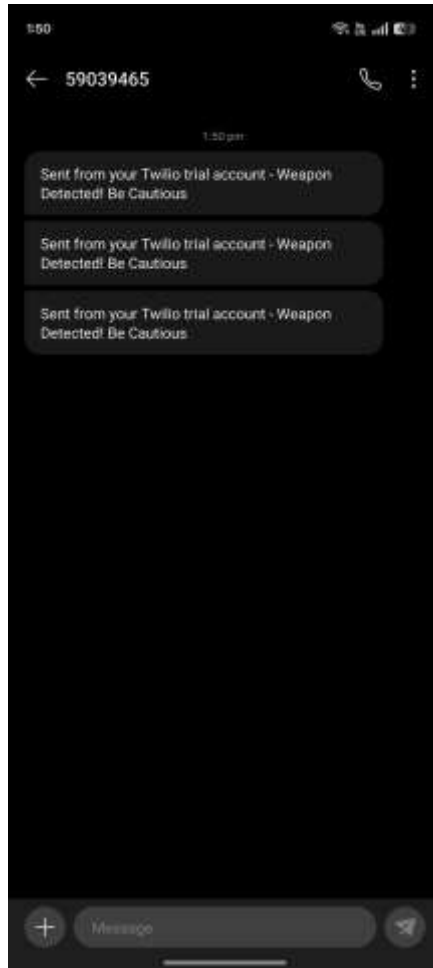
Test Case ID	Integration Scenario	Test Description	Expected Output	Result
ITP-01	Detection + Alert System	Frame triggers detection → sends email/SMS	Alert sent to user	Pass
ITP-02	Client-Side Feed to Server	Webcam stream is posted to Django server	Frame received and processed	Pass
ITP-03	Server Response + Log	Detected frames are saved and logs are updated	Detection logged in database	Pass

RESULTS

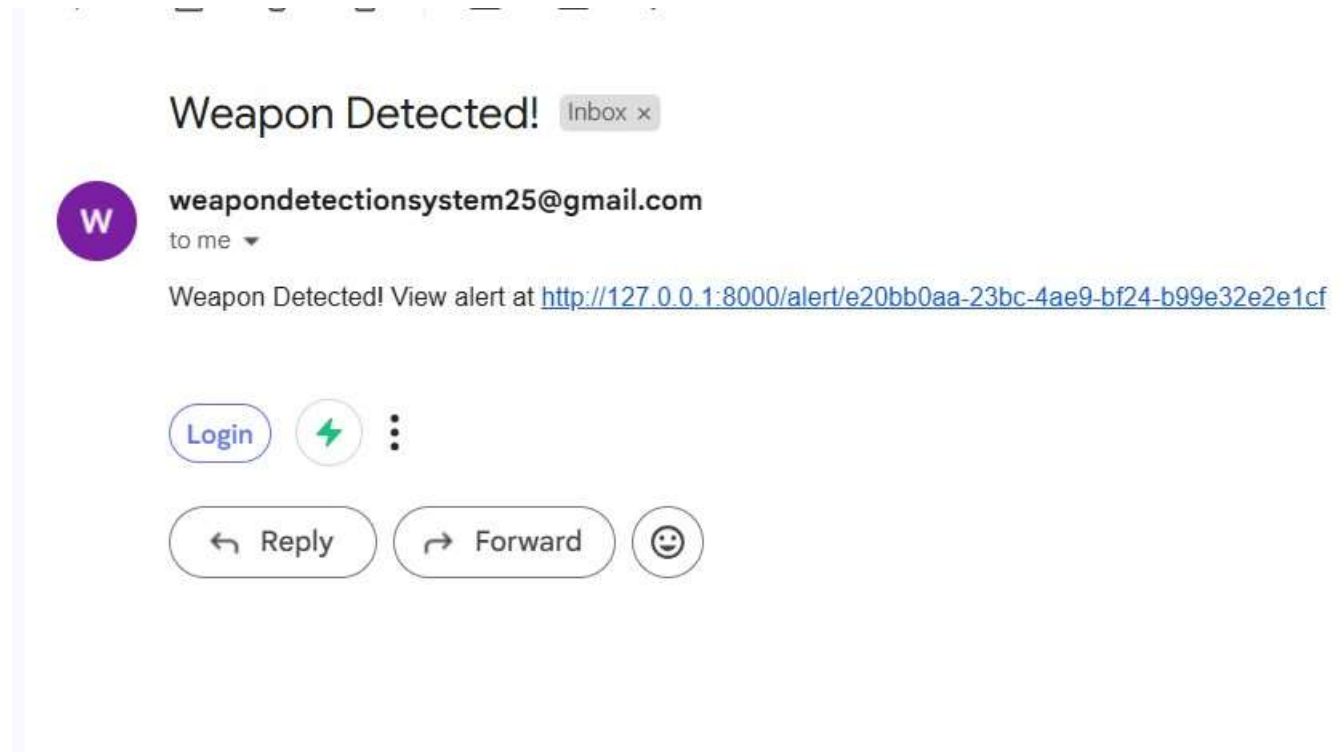


Weapon Detected

RESULTS



SMS Alert



Email Alert

CONCLUSION

Summary:

- The project develops a Real-Time Weapon Detection System using YOLOv8, OpenCV, Django.
- Stores detected frames, and instant alerts are sent for rapid security responses.
- The system aims to improve public safety and security through accurate and timely weapon detection.

Expected Outcomes:

- Real-time detection of weapons with high accuracy.
 - Instant alerts for security personnel.
 - Scalable deployment and efficient data management.
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CONCLUSION

Future Scope:

- Expand detection to other threats (e.g., **explosives**).
- Improve **adaptability** for different environments.
- Smart city and public safety integration



○ **AI – DRIVEN SMART SURVEILLANCE FOR WEAPON DETECTION USING YOLOv8**

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GitHub Repository
<https://github.com/bharathraj1024/AI-Driven-Smart-Surveillance-for-Weapon-Detection-Using-YOLOv8>

Demo Video
https://youtu.be/vkhxaQHgLU0?si=3aFgimUsW_8EpK3y

Portfolio
<https://react-portfolio-ochre-gamma.vercel.app/>

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